

When Does Forecast Accuracy Predict Inventory Cost?

Evidence from a Maritime Spare-Parts Fleet

Nishchay Patel

September 7, 2026

Abstract

Forecasters are usually chosen on point accuracy, but accuracy is not what an inventory operation pays for. Theodorou, Spiliotis and Assimakopoulos (2025) showed on M5 retail data that the link between accuracy and inventory cost is conditional on the cost structure: accuracy matters when holding cost is comparable to or larger than the cost of a stockout, and much less when stockouts dominate. We test that boundary out of domain, on 15,348 SKUs of real maritime spare-parts data from a multi-brand cruise group, under an (s, S) policy rather than order-up-to, with the operator’s own deployed policy as a baseline. The boundary is visible: the fleet-level rank correlation between mean absolute error and simulated cost is +0.591 where holding cost dominates, with a bootstrap interval of $[+0.49, +1.00]$ that excludes zero ($B = 20,000$; the bootstrap puts 0.14 % of replicates at or below zero), while where stockout cost dominates the point estimate is -0.345 with an interval of $[-0.76, +0.44]$ that does not. Because 96.8 % of these parts sit in the stockout-dominated regime the pooled correlation is indistinguishable from zero (+0.018). Carrying the sampling design weights, which reach 96 : 1 because the sample is equal-allocation, sharpens the contrast rather than dissolving it (+0.991 against -0.882), but only the positive side survives dropping the one stratum that is 68 % of the fleet. The sign change is not evidence that accuracy predicts cost. MAE is minimised at the median of realised demand, which is zero for 77.3 % of these SKUs, and on such a series MAE is provably monotone in the forecast, so ranking forecasters by MAE is ranking them by how much they forecast: the per-SKU rank correlation between level and MAE has median +1.000. The regime boundary itself follows analytically. With underage cost C_u and overage cost C_o the cost-minimising forecast is the quantile at the critical fractile $C_u/(C_u + C_o)$, which equals the 0.5 fractile MAE targets at exactly one unit price, here \$1,600, and that is precisely the observed crossover. We test the obvious remedy and it fails: scoring at each SKU’s critical fractile is confounded with level just as badly in the opposite direction. The confound is escapable only under a policy that consumes the whole predictive distribution, and then only by scoring at the fractile the decision actually reads, since a generic distributional score performs worse than MAE. Finally we test generality on the public M5 data and it does not transfer: the analytic condition holds for 10.1 % of M5 series against 77.3 % here, and the confound is correspondingly mild. The severity is therefore a measurable property of the evaluation window rather than of intermittency, and that reconciles why competition-data and spare-parts studies disagree. Two secondary results: a zero-shot Chronos-T5 model is the most accurate and the cheapest forecaster in the set, and ten of the eleven model-driven policies beat the deployed parameterisation on cost, though our simulator is not validated against realised stockouts so we report that direction without its magnitude. We also report a negative result. Scoring against right-censored quarters, which is a real feature of this data, does not reorder the forecasters at all. The anonymised data and the full pipeline are released.

Keywords: intermittent demand; spare-parts inventory; cost-aware forecast evaluation; cost regime; foundation models; maritime operations.

1 Introduction

Forecasting models are judged by accuracy metrics such as MAE, MASE and RMSSE. In spare-parts management the model is not the end of the line: it feeds an (s, S) inventory policy, and what that policy costs is holding cost plus whatever stockouts cost. Whether the two criteria agree has been asked for two decades (Syntetos & Boylan 2006; Teunter et al. 2010; Prak & Teunter 2019; Kolassa 2016), and the answers have not been consistent.

Recent work explains part of that inconsistency. Petropoulos, Wang & Disney (2019) compared inventory performance across the M3 methods, and Theodorou, Spiliotis & Assimakopoulos (2025) went further on the M5 data: they find the accuracy-to-cost link is weak and conditional, and specifically that accuracy is more relevant when holding cost is similar to or larger than lost-sales cost. Where lost-sales cost dominates, they note, the preferable method may not be the most accurate one, and they single out intermittent products.

That is a testable prediction, and it has not been tested outside retail competition data. We test it on the spare-parts operation of a multi-brand cruise group: three brands, 40 vessels, around 220k SKUs with quarterly forecast history. Our setting differs from theirs in three ways that matter. The demand is far more intermittent than M5 retail. The policy is (s, S) rather than order-up-to. And the baseline is not another model but the (s, S) parameters the operator actually runs.

Contributions.

- **MAE is confounded with forecast level, by identity rather than by tendency.** For a flat forecast, MAE is minimised at the median of realised demand. That median is zero for 77.3 % of these SKUs, and on such a SKU MAE is strictly increasing in the forecast, so MAE-rank is level-rank inverted. Per SKU across the eleven forecasters the level-MAE rank correlation has median +1.000, with 78.6 % of SKUs at exactly +1.
- **The regime boundary has an exact derivation.** The cost-minimising forecast is the quantile at the newsvendor critical fractile $q^* = C_u / (C_u + C_o)$. Here $q^* = 0.5$ exactly at a unit price of \$1,600, which is the empirically observed crossover. MAE implicitly targets 0.5 for every SKU regardless of price, so it is aligned with the cost structure at one price and points the wrong way on both sides of it. That is why the correlation reverses sign rather than merely weakening.
- **The obvious remedy fails, and the working one is narrower than expected.** Scoring at each SKU's own q^* is confounded with level just as badly in the opposite direction (per-SKU pinball-level rank correlation median -1.000). Under a policy that consumes the whole predictive distribution the confound is escapable, but only by scoring at the fractile the decision reads: pinball at the service level predicts cost better than MAE (median +0.800 against +0.632) while a generic five-quantile CRPS does worse (+0.400).
- **The severity is measurable in advance, and does not generalise.** On the public M5 data the same statistic has median +0.194 with 1.5 % of series at +1, because the analytic condition holds for only 10.1 % of M5 series against 77.3 % here. Conditioning on it, the mechanism does reappear (median +0.614 where the condition is met against +0.157 where it is not). So the confound is a property of how zero-dense the evaluation window is, not of intermittent demand as such.
- **The level of aggregation also changes the answer.** The within-SKU rank correlation is

+0.387 while the fleet-level correlation across forecasters, which is what a practitioner choosing one model implicitly uses, is +0.018. These are different estimands rather than contradictory results.

- **A zero-shot foundation model is simultaneously the most accurate and the cheapest forecaster** in a field of eleven that includes the operator’s deployed methods and the standard Croston family.
- **A negative result on censoring.** The most recent quarters of this feed are incomplete. Holding the training window and the horizon fixed and moving only the test window into the censored region leaves the cost ranking unchanged. Censoring is real in the data and consequential for reported demand, but it does not reorder forecasters.

Section 2 places the work; Section 3 covers data; Section 4 methods; Section 5 results; Sections 6 to 8 discuss, bound and conclude. A reproducibility note records the analysis history, including several corrections.

2 Related work

Intermittent-demand forecasting. Croston (1972) smoothed demand size and inter-demand interval separately. Syntetos & Boylan (2005) corrected its inversion bias, giving SBA; Teunter, Syntetos & Babai (2011) smoothed the demand probability instead (TSB); Nikolopoulos et al. (2011) introduced temporal aggregation (ADIDA/IMAPA); Kourentzes (2014) added parameter tuning. MAPE is undefined on mostly-zero series and plain MAE rewards a flat zero forecast, so Hyndman & Koehler (2006) standardised the scaled error (MASE).

Accuracy versus inventory performance. Syntetos & Boylan (2006) showed empirically that accuracy and stock-control rankings can diverge; Teunter et al. (2010) and Prak & Teunter (2019) developed the gap analytically; Kolassa (2016) argued point-error metrics are the wrong target for count data. Petropoulos, Wang & Disney (2019) measured inventory performance across M3 methods. Theodorou, Spiliotis & Assimakopoulos (2025) supply the result this paper builds on: using the M5 data, quantile forecasts and an order-up-to policy, they find the accuracy-cost relationship depends on the balance between holding and lost-sales cost. Our contribution is to test that dependence on real operational data, in a much more intermittent regime, under a different policy class, against a deployed baseline.

Probabilistic forecasting and calibration. Gneiting et al. (2007) formalised calibration and sharpness. We use predictive quantiles to set safety stock and report the coverage they achieve.

Global and foundation-model forecasting. Cross-learning across series (Salinas et al. 2020; Bandara et al. 2020; Makridakis et al. 2022) set up the zero-shot foundation models, of which Chronos (Ansari et al. 2024) is one. Vandeput (2020) cautions that machine learners over-predict when most observations are zero. Head-to-head evaluations of a foundation model against Croston-family methods on intermittent spares, scored by inventory cost rather than accuracy alone, appear to be scarce.

Inventory policy. The (s, S) machinery is textbook (Silver, Pyke & Thomas 2017; Snyder & Shen 2019).

3 Data and deployed pipeline

3.1 Source, scope and sampling

The operator runs four maintenance-management instances, three legacy and one migrated. We extract spare-part consumption, the parts master, the forecast-analysis table and the deployed (s, S) parameter table, restricted to the cruise group’s operating segment. Anonymisation is keyed and salted: every identifying field (brand, vessel, SKU, part type, vendor, department) becomes a namespaced surrogate, consistent across tables. Free-text fields are dropped. Dates, quantities, prices, lead times, the demand-pattern segmentation, ABC class and criticality are preserved.

The analysis runs on a **stratified sample of 15,348 SKUs** drawn from the 219,789-SKU forecast panel by equal allocation across demand-pattern and ABC cells, with a per-cell cap of 1,579. Equal allocation buys precision in the small, high-value cells that a proportional sample would barely reach, and it is the right design for a paired model comparison. It is the wrong design for reading a raw mean as a fleet quantity: sampling fractions run from 1.0 % to 100 %, so the design weight $w_h = N_h/n_h$ spans 1.00 to 96.12, and the single Intermittent $\times$ C cell is 69 % of the panel against 10 % of the sample.

We therefore report both. Model comparisons, which are paired within SKU, use the sample unweighted. Every quantity stated as a property of the fleet is also reported as a Horvitz–Thompson estimate carrying w_h , with a bootstrap that resamples within strata (Section 5.9). Kish’s effective sample size under those weights is 3,127, not 15,348, and we quote it wherever a weighted interval is given.

3.2 Right-censoring and the evaluation window

Each SKU carries 28 quarterly observations. The fraction with nonzero demand holds at about 5 to 7 % through panel quarter 20, then falls to roughly 1 % by quarters 25 to 28 (Figure 1). This is a reporting lag rather than a demand collapse: the consumption log is stable through the most recent complete calendar quarter and only then drops, because recent transactions have not all been entered. The last panel quarters are incomplete.

We therefore evaluate on **train = panel quarters 1 to 16, test = quarters 17 to 20**, four quarters at roughly 5 % nonzero, matching the training level. Of the 15,348 SKUs, 5,929 (38.6 %) see positive demand in that window. Section 5.6 reports what happens when the test window is deliberately placed in the incomplete region instead.

4 Methods

4.1 Forecasters

Eleven models in three tiers. **Classical (seven)**: the operator’s deployed SBA, SES and two-quarter moving average, plus Croston without bias correction, TSB, ADIDA and IMAPA. **Distributional (two)**: an intercept-only Zero-Inflated Poisson (ZIP) and a Hurdle Negative-Binomial (HNB), fit by method-of-moments and turned into predictive quantiles by sampling 1,000 draws. **Global and foundation (two)**: a global LightGBM (Ke et al. 2017) with separate occurrence, size and quantile heads, and Chronos-T5-small run zero-shot on the 16-quarter context with 1,000 seeded sample paths.

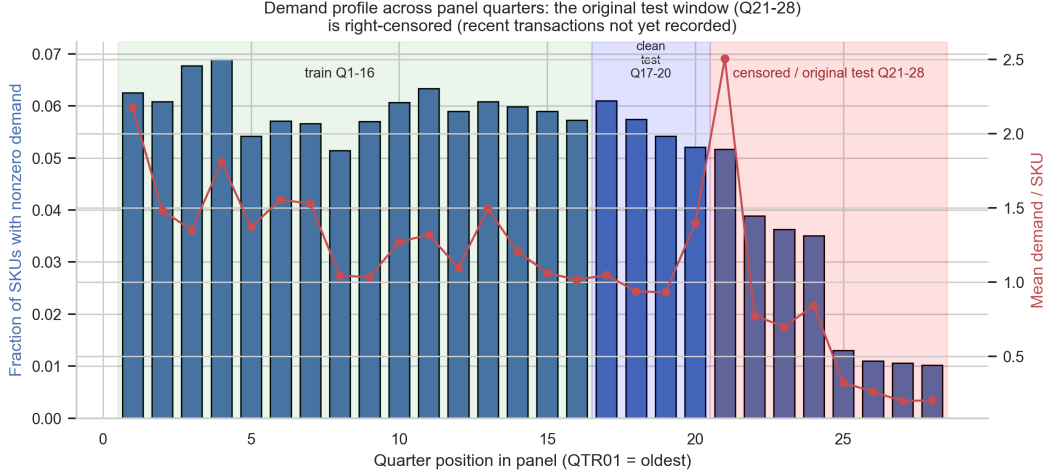


Figure 1: Demand profile across panel quarters. The most recent quarters are incomplete; the evaluation window is Q17–20.

Two implementation points matter for reproducibility. LightGBM’s quantile heads are fit on positive-target rows, so they estimate the size quantile conditional on occurrence; the predictive quantile of demand is the hurdle mixture of that with the occurrence probability, not the conditional quantile gated by an indicator. And LightGBM is non-deterministic across runs when multithreaded unless its seeds, `deterministic`, `force_row_wise` and thread count are all pinned. Both points are elaborated in the reproducibility note, because getting either wrong changes the conclusions.

4.2 (s, S) cost simulator

For each SKU and forecaster we build an (s, S) policy and replay the test-window demand: quarterly review, integer-quarter lead times ($\lceil \text{days}/91.3125 \rceil$, minimum one), and holding cost charged on end-of-quarter positive on-hand. Default configuration: holding 25 %/yr of unit value, ordering \$50 per order, stockout \$100 per unit short. Section 5.5 sweeps all three.

Four modelling choices shape these results and are stated explicitly, because each one is consequential on a four-quarter window.

- **Unmet demand is lost, not backordered.** A stockout is charged once per unit short and the shortfall does not carry into the next quarter.
- **Inventory starts saturated** at on-hand = S . On this window that is not a neutral choice: 83.0 % of SKUs never place an order in the four test quarters, so for most of the sample the simulation measures the cost of holding an initial position rather than the cost of replenishing. Starting from zero instead raises mean cost by about 5 % but drops mean fill from 0.965 to 0.828, so the start condition matters far more for service than for cost.
- **Fill rate is 1.0 by definition when a SKU has no realised demand**, which is 61.4 % of the sample. Reported fleet fill rates are therefore mostly definitional; the demand-active mean fill is closer to 0.91. We use fill only to match service between the two policy formulas, never as a headline result.
- **Cost is dominated by holding.** Under the default configuration the split is 90.5 % holding, 9.0 % stockout, 0.5 % ordering. Since holding cost is close to monotone in forecast level, the

cost ranking is substantially a ranking of forecast bias, and Section 5.3 should be read with that in mind.

The service level enters only through $z = \Phi^{-1}(\text{SL})$ in the safety-stock term, so it is applied as a cycle-service (Type-1) quantity even though the criticality tiers were specified as availability targets.

Normal-approximation policy. The reorder point is

$$s = \bar{D} \cdot \text{LT} + z \sqrt{\text{LT} \cdot \sigma_D^2 + \bar{D}^2 \sigma_{\text{LT}}^2}, \quad S = s + \text{EOQ},$$

with EOQ from the standard formula. σ_D is held model-agnostic, one training-window standard deviation shared by every forecaster, so the only thing that varies across forecasters is the point forecast $\bar{D}$. This isolates point-forecast quality, at the cost of not testing forecaster-specific uncertainty (Section 7).

Native-quantile policy. For the four models that emit quantiles we set s from the model’s own predictive quantile of lead-time demand at the SKU’s criticality-tiered service level, then rescale by a bisection-searched multiplier κ until fleet fill matches that model’s normal-approximation policy, and compare cost at equal service.

4.3 Metrics and the cost regime

Point accuracy is MAE, with RMSSE as a genuinely independent check. We do not report MASE as a robustness check on the rank statistic: MASE is a strictly positive per-SKU rescaling of MAE, so the per-SKU rank correlation is identical to the MAE one by construction and corroborates nothing.

We report the accuracy-cost relationship at two levels, because they answer different questions. The **within-SKU** statistic is the mean per-SKU Kendall’s τ between MAE-rank and cost-rank across the eleven forecasters, with SKU-level bootstrap CIs. The **fleet-level** statistic is the Spearman correlation between each forecaster’s mean MAE and its mean cost, across the eleven forecasters. The second is what a practitioner implicitly uses when choosing one model for the whole fleet.

Cost regime. Under the default configuration, per-unit-quarter holding cost is $(0.25/4) \cdot \text{price}$ and the stockout penalty is a flat \$100, so the two are equal at a unit price of \$1,600. SKUs below that are stockout-dominated; above it, holding-dominated. This gives a direct test of the Theodorou et al. boundary.

5 Results

All results are on the uncensored window: train quarters 1 to 16, test 17 to 20.

5.1 Accuracy

Mean test MAE, best to worst: Chronos 6.525, MA 7.598, SES 7.889, IMAPA 8.201, ADIDA 8.311, HNB 8.595, ZIP 8.595, TSB 8.633, LGBM 8.775, SBA 8.948, Croston 9.107 (Figure 2, x-axis). Chronos wins zero-shot with no fine-tuning. SBA and Croston come last, which fits: a bias-corrected estimator is not built to minimise raw error when most periods are zero.

ZIP and HNB have algebraically identical point forecasts here (both reduce to the training sample mean), so the study contains ten distinct point forecasters, not eleven. We keep both because their predictive distributions differ, which matters in Section 5.4.

5.2 Cost

Mean cost per SKU, cheapest to dearest: Chronos \$2,097, HNB \$2,554, ZIP \$2,555, TSB \$2,564, SBA \$2,581, Croston \$2,585, IMAPA \$2,607, SES \$2,657, ADIDA \$2,691, MA \$2,879, LGBM \$2,936, at fill rates from 0.956 to 0.980 (Figure 2, y-axis).

Chronos is both the most accurate and the cheapest. The second-most-accurate model, the moving average, is the second most expensive. That pair is the whole problem this paper is about, and Section 5.3 explains why it happens.

The ordering is almost entirely unresolvable, and we quantify that rather than noting it in passing. Simulated cost is extremely right-skewed: 3.2 % of SKUs carry 82.3 % of it. Those are not an independently identified tail; they are the holding-dominated group of Section 5.3, since the same unit-price threshold defines both. A paired bootstrap over SKUs ($B = 20,000$) gives Chronos a mean of \$2,097 with a 95 % interval of [\$543, \$5,050]. On the paired difference against the cheapest model, **only one of ten models is distinguishable**: LightGBM, at +\$839 [+219, +1,806]. Chronos against Hurdle-NB is +\$457 [−231, +1,605]. So Chronos is the cheapest *point estimate* but is not statistically separable from nine of the other ten, and no claim in this paper rests on the cost ordering of adjacent models.

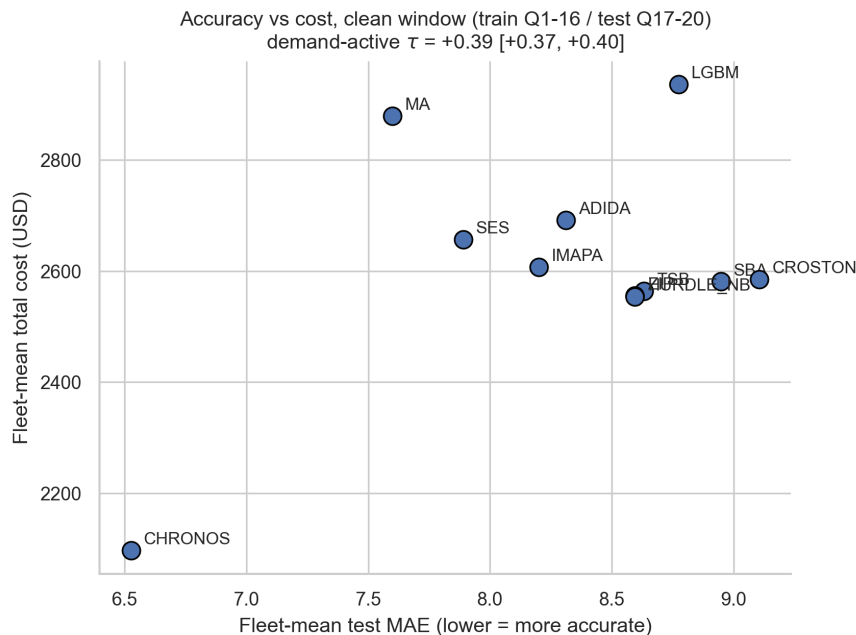


Figure 2: Accuracy (test MAE) versus simulated cost. The most accurate model (Chronos) is also the cheapest; the second (MA) is the dearest.

5.3 The accuracy-cost relationship depends on regime and on aggregation

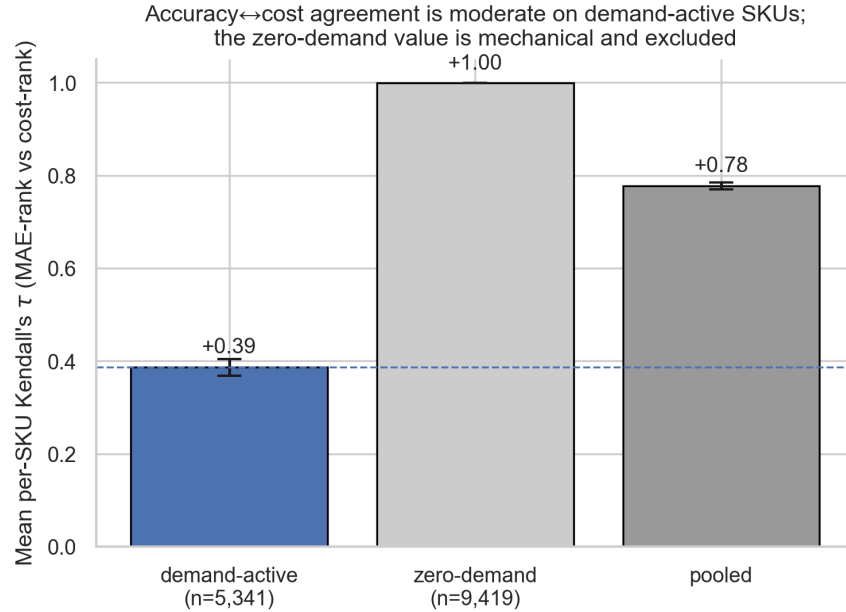
Within-SKU. On a SKU with no realised demand both MAE and holding cost increase with forecast magnitude, so τ is positive by construction; we rely on the demand-active value. Under

Table 1: Mean per-SKU Kendall’s τ between MAE-rank and cost-rank.

SKU subset	n (τ defined)	τ	95 % CI
Demand-active	5,341	+0.387	[+0.369, +0.405]
Zero-demand	9,419	+0.999	mechanical, excluded
Pooled	14,760	+0.778	inflated by zero-demand

RMSSE the same statistic falls to +0.175 [+0.154, +0.192], so even within-SKU the answer depends on the error metric (Figure 3).

Fleet-level. Across the eleven forecasters, Spearman ρ between mean MAE and mean cost is **+0.018** ($p = 0.958$): no relationship at all. The within-SKU and fleet-level statistics are both correct and they disagree, because they ask different questions. Averaging ranks within SKUs measures whether a better forecast for a given part tends to cost less for that part. Correlating model means measures whether a more accurate model is a cheaper model for the fleet, which is the question a practitioner selecting one forecaster is actually asking.

**Figure 3:** Mean per-SKU Kendall’s τ (MAE-rank versus cost-rank), split by demand activity. The zero-demand value is mechanical and excluded.

By cost regime. Splitting at the \$1,600 crossover:

Table 2: The accuracy-cost relationship by cost regime.

Regime	n SKUs	fleet ρ	within-SKU τ	most accurate	cheapest
Stockout-dominated ($< \$1,600$)	14,861	−0.345	+0.385	Chronos	LGBM
Holding-dominated ($\geq \$1,600$)	487	+0.591	+0.491	Chronos	Chronos

A paired bootstrap over SKUs puts intervals on these. The holding-dominated correlation is +0.591 with a 95 % interval of $[\mathbf{+0.49}, \mathbf{+1.00}]$, which excludes zero, so it survives inference that resamples SKUs rather than relying on an eleven-point asymptotic test (whose p was 0.056). The stockout-dominated correlation is -0.345 with an interval of $[\mathbf{-0.76}, \mathbf{+0.44}]$, which includes zero: there is no detectable relationship there, and we do not claim a negative one. Pooled across all SKUs the interval is $[-0.29, +0.75]$, also including zero.

We report these intervals to two decimals deliberately. At $B = 2,000$ the holding-dominated lower bound was not stable in the third decimal, ranging from +0.409 to +0.545 across five seeds; at $B = 20,000$, which is what we now run, the same sweep spans +0.464 to +0.500. The percentile endpoint of a rank correlation over eleven model means is the fragile statistic here, so alongside it we report the bootstrap share of replicates at or below zero, which is stable: 0.0014 in the holding-dominated group, 0.727 in the stockout-dominated group and 0.447 pooled. This is the Theodorou et al. prediction, reproduced on independent operational data under a different policy class. Where holding cost dominates, accuracy predicts cost and the most accurate model is the cheapest. Where stockout cost dominates, the correlation is negative and the most accurate model is not the cheapest. Because 96.8 % of these parts are stockout-dominated, the pooled fleet correlation is nil, and any study of intermittent spares that reports a single pooled number will conclude accuracy does not matter, while a study of high-value slow movers will conclude it does.

Why the sign changes: MAE is confounded with forecast level. The regime pattern above is not evidence that better forecasts cost less.

Start analytically. For a flat forecast $f > 0$ on a window $d_1, \dots, d_H$,

$$\frac{d\text{MAE}}{df} = \frac{\#\{t : f > d_t\} - \#\{t : f < d_t\}}{H},$$

so if more than half the d_t are zero then MAE is strictly increasing in f for every $f > 0$. On such a SKU, ranking forecasters by MAE is *identically* ranking them by forecast level, inverted. This is a property of the metric on zero-dense data, not a tendency that washes out in a larger sample.

The condition binds here. The median of the test window is zero for **77.3 %** of SKUs (61.4 % are entirely zero), and even among demand-active SKUs 41.3 % still have a zero median. Measured per SKU across the eleven forecasters, the rank correlation between forecast level and MAE has **median** +1.000, with 78.6 % of SKUs at exactly +1 ($n = 14,760$). These use 15,348 SKUs rather than eleven model means.

Table 3: Decomposition of the accuracy-cost relationship. In the holding-dominated group MAE and forecast level are perfectly rank-correlated, so nothing survives conditioning on level.

Subset	$\rho(\text{MAE, cost})$	$\rho(\text{MAE, level})$	partial	$\rho(\text{MAE, hold})$	$\rho(\text{MAE, stockout})$
All	+0.018	+0.982	-0.187	+0.018	-0.509
Stockout-dominated	-0.345	+0.964	+0.548	+0.827	-0.509
Holding-dominated	+0.591	+1.000	+0.000	+0.591	+0.091

In the holding-dominated group, where the raw correlation is strongest, MAE and forecast level are perfectly rank-correlated and the partial correlation is exactly zero: accuracy and level cannot be separated there at all. The mechanism is visible in the last two columns. Forecast level raises holding cost ($\rho = +0.827$, $p = 0.002$) and lowers stockout cost ($\rho = -0.509$). Which of those two

components dominates the total is exactly what the cost regime decides, and that, not forecast quality, is what sets the sign of the accuracy-cost relationship.

The practical consequence is stronger than “accuracy is a weak proxy”. Because the confound is structural rather than incidental, MAE cannot serve as a model-selection criterion for these SKUs in *either* regime: where it appears to work it is standing in for level, and where it appears to fail it is also standing in for level, with the sign flipped by the cost structure.

5.4 Why the boundary sits where it does, and what to use instead

The crossover is derivable. In a newsvendor trade-off with underage cost C_u and overage cost C_o , the cost-minimising order quantity is the demand quantile at the critical fractile $q^* = C_u / (C_u + C_o)$. Here C_u is the \$100 stockout penalty and $C_o = (0.25/4) \times \text{price per quarter}$, so

$$q^*(\text{price}) = \frac{100}{100 + 0.0625 \text{ price}},$$

which equals 0.5 at exactly **price = \$1,600**: the regime crossover of Section 5.3, derived rather than observed. MAE implicitly targets the 0.5 fractile for every SKU regardless of price, so it is aligned with the cost structure at exactly one price and misaligned on both sides, which is why the correlation reverses sign instead of merely weakening. The misalignment is severe: median q^* here is **0.981** and 96.8 % of SKUs require $q^* > 0.5$.

The obvious remedy does not work. If MAE targets the wrong fractile, score each SKU at its own q^* instead. We tested that and it fails. Pinball loss at q^* is confounded with level just as badly in the opposite direction: the per-SKU rank correlation between that loss and forecast level has median **−1.000**, and fleet-level $\rho(\text{pinball}, \text{cost})$ is **−0.491** against MAE’s +0.018. It helps only where holding dominates (+0.809, $p = 0.003$). Swapping the metric swaps the direction of the confound rather than removing it.

The reason is structural. Under the model-agnostic σ the policy sees the forecast only through the scalar $\bar{D}$, so simulated cost is a deterministic function of level and *no* point-error metric can add information about cost. There is no second channel to exploit.

A distributional policy does open one, but only just. Under the native-quantile policy the reorder point is $s = \text{LT} \cdot Q_\alpha$, so the policy reads a quantile rather than a mean. That is a genuine second channel: the per-SKU rank correlation between level and Q_α has median +0.738, with $|\rho| > 0.99$ on only 14.4 % of SKUs, and cost tracks Q_α (median $\rho = +1.000$) more tightly than level (+0.816).

Table 4: Which metric predicts native-quantile cost (per-SKU median rank correlation, demand-active SKUs).

Metric	median ρ with cost
MAE	+0.632
Pinball at the service level	+0.800
CRPS over the five fitted quantiles	+0.400

The ordering is the useful result, and it is not “distributional beats point”. A generic distributional score is *worse* than MAE, because CRPS averages over the whole quantile grid including the median

region the decision never reads. What wins is the loss evaluated at the specific fractile the policy consumes. Only four models emit quantiles, so the fleet-level version of this comparison rests on four points and we do not interpret it.

5.5 Does the confound generalise? A test on public M5 data

If the confound were a property of intermittent demand it would reappear on the M5 competition data. It does not. Across 30,490 M5 series and eight classical forecasters, the per-series rank correlation between forecast level and MAE has median **+0.194**, with 1.5 % of series at exactly +1, against median +1.000 and 78.6 % here. By demand class the M5 medians are ordered as expected (Smooth +0.157, Erratic +0.229, Intermittent +0.253, Lumpy +0.301) but none is close to the maritime result.

The analytic condition explains the gap. Median demand is zero over the out-of-sample window for **10.1 %** of M5 series against **77.3 %** here; at a matched four-period horizon M5 is still only 5.2 %. So the difference is how zero-dense the evaluation window is, not the horizon length and not the Syntetos-Boylan class label. Conditioning on the analytic condition, the mechanism does reappear on M5: median rank correlation **+0.614** on the 3,082 series where median demand is zero, against **+0.157** on the 27,384 where it is not.

The claim is therefore conditional and its severity is measurable in advance from the evaluation window alone. That also reconciles a disagreement rather than adding to it: competition-data studies and spare-parts studies sit on opposite sides of this condition.

5.6 Predictive-quantile policies

Table 5: Fill-matched cost change of the native-quantile policy against the normal-approximation policy, with achieved 99th-percentile coverage.

Model	κ to match fill	Q99 coverage	cost Δ at matched fill
ZIP	0.88	0.963	−14.8 %
HNB	0.91	0.973	+1.5 %
LGBM	0.51	0.995	−64.4 %
Chronos	3.83	0.961	+4.2 %

All four reach the target fill. The two models whose upper tails are best calibrated at the nominal 99 % level, LightGBM (0.995) and HNB (0.973), sit at opposite ends of the cost outcome (Figure 4, Figure 5), so calibration at a single level does not by itself determine whether a predictive-quantile policy pays. The ZIP saving is stable across sampling seeds (mean −15.2 %, range −15.5 % to −14.8 % over five seeds). A global multiplicative conformal correction moves every one of these by less than half a percentage point, so it neither rescues nor damages any model here.

We make no general claim from this table. The honest summary is that a predictive-quantile policy helped two of four models on this data and was approximately neutral for the other two, and that we cannot separate the effect of tail calibration from the effect of the fill-matching procedure itself.

5.7 Deployed policy comparison

On the 1,579 SKUs where the deployed parameter set can be matched, ten of the eleven model-driven policies beat the deployed policy on cost. The best, HNB, costs \$567 against \$1,151, a 51 % reduction

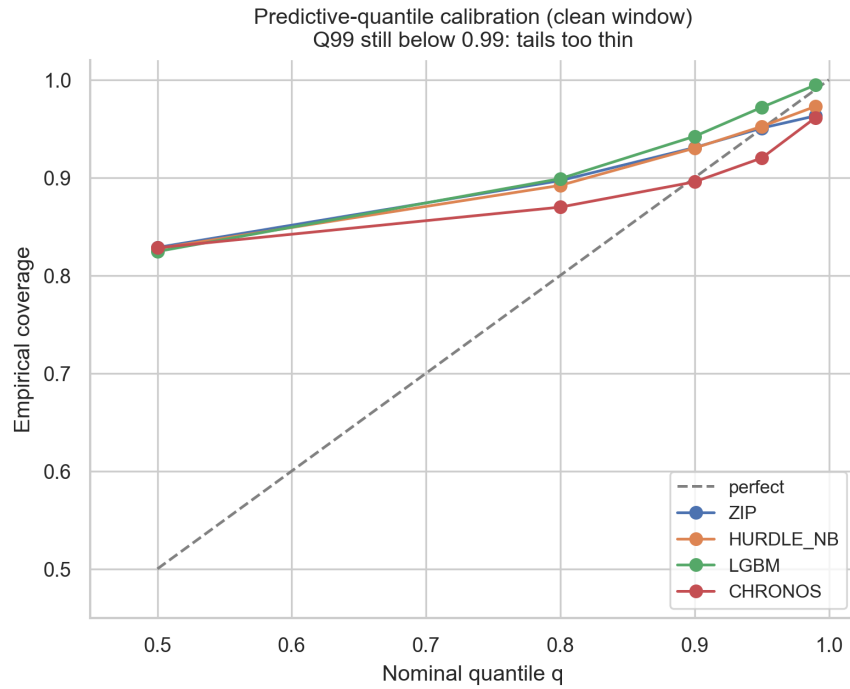


Figure 4: Predictive-quantile calibration on the evaluation window.

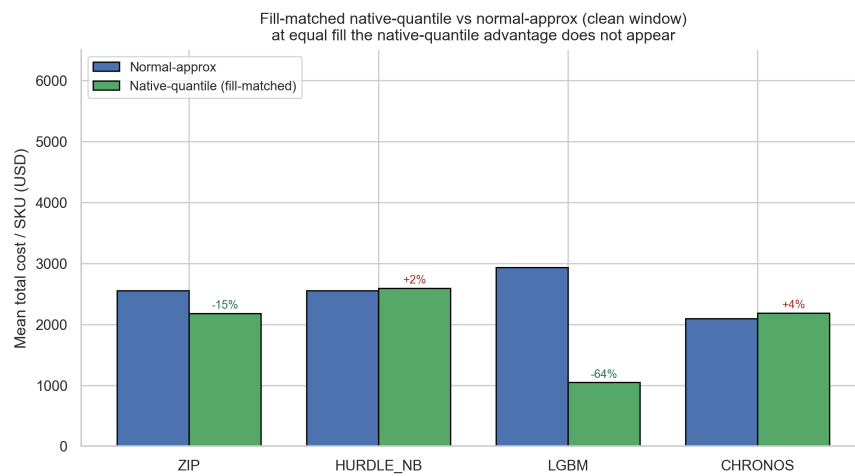


Figure 5: Fill-matched native-quantile versus normal-approximation cost.

(Figure 6). LightGBM is the exception at \$1,543, or 34 % more expensive than deployed, because its point forecast over-stocks and this overlap is richer in demand-active SKUs than the sample as a whole.

The direction is robust and now has an interval: a paired bootstrap over the overlap SKUs puts the gap at -50.7% with a 95 % interval of $[-60.4\%, -41.0\%]$, comfortably excluding zero. The cheapest model in each cell beats deployed in 80 of 80 cost-configuration cells and in all four service-level scenarios, by 49 % to 53 %. Which model is cheapest is configuration-dependent (IMAPA wins 31 cells, Chronos 23, HNB 13, ZIP 9, ADIDA and SBA 2 each), so there is no single best forecaster to crown.

We do not report the magnitude with confidence. The data contain no realised stockouts, so the simulator cannot be validated against ground truth. Against the recorded inventory snapshot it tracks in shape but not in level: simulated deployed on-hand correlates with recorded stock at $+0.663$ in logs but sits at about half its level (median ratio 0.50), and the simulated deployed fill of 0.864 falls short of the operator’s stated 0.95 to 0.99 goal. The simulator under-stocks relative to reality. Model-versus-model comparisons share the simulator and are unaffected; the absolute deployed figures are directional. The overlap is also a biased subsample, over-weighted toward lumpy and high-ABC parts. We do not compare fill on the overlap, because model fill was computed sample-wide rather than on this subset.

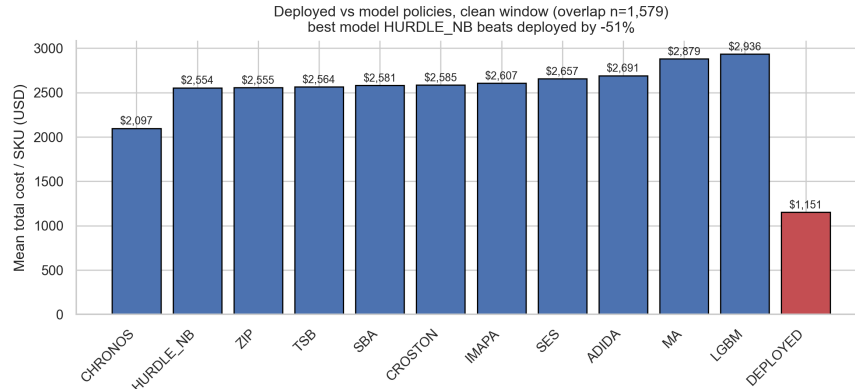


Figure 6: Deployed versus model-driven policy cost on the overlap SKUs.

5.8 A negative result: censoring does not reorder forecasters

Because the recent quarters are incomplete, an obvious worry is that a test window placed there would flatter models that under-forecast. We tested it. Holding the training window (quarters 1 to 16) and the horizon (four quarters) fixed, and moving only the test window into the incomplete region (quarters 25 to 28), the demand-active fraction falls from 38.6 % to 11.4 %, confirming the censoring is real and severe. The cost ranking does not move: the ordering of all ten models is identical across the two windows, the moving average is ninth of ten in both, and the within-SKU τ is unchanged ($+0.404$ clean against $+0.419$ censored, overlapping intervals).

A ranking change does appear if one compares the clean window against the original naive split (train quarters 1 to 20, test 21 to 28), where the moving average is the cheapest model. But that comparison moves the training window and doubles the horizon as well as shifting the test period, so it cannot be attributed to censoring. We report this because we initially believed the opposite,

and because the controlled version of the experiment is cheap and we suspect other cost-aware studies have not run it.

5.9 Two design checks: information sets and design weights

Two things about the setup would not survive a referee. Three of the eleven forecasters saw less history than the rest, and the sample is not the fleet whose means the paper reports. We fixed both and report what moved.

Equal information sets. Croston, SBA and TSB update their smoothed state from the previous observation, so reading the forecast at the last training column returns a forecast whose state has consumed only fifteen of the sixteen training quarters. The final quarter reaches it only through the initialisation constants, whose weight decays as $(1 - \alpha)^T$. Every other forecaster here conditions on the full window: SES takes an explicit extra step, the moving average reads the last two quarters directly, ADIDA and IMAPA end with a smoothing step on the most recent aggregated block, ZIP and Hurdle-NB fit moments over all sixteen columns, and LightGBM and Chronos take lags and context ending at the last quarter. Three of eleven models were being scored on a shorter history than their competitors.

Running one further iteration of the same recurrence removes the asymmetry. The three affected models improve, by 0.36 %, 0.37 % and 2.12 % of MAE and 0.30 % to 0.56 % of cost, and the other eight are bit-identical, which is the check that the change touched nothing else. No conclusion moves. Chronos remains both the most accurate and the cheapest; the holding-dominated correlation is unchanged at +0.591; the within-SKU τ is +0.387 either way; the deployed gap moves from -50.7 % to -50.9 %. Two labels do change: TSB rises from fourth to second cheapest, and it replaces Hurdle-NB as the cheapest model on the deployed overlap. Since Section 5.2 already establishes that adjacent models are not separable, neither label carries an inferential claim.

Design weights. Sampling fractions run from 1.0 % to 100 %, so the Horvitz–Thompson estimate carrying $w_h = N_h/n_h$ answers a different question from the raw mean: what the fleet looks like, rather than what the sample does. The two differ substantially.

Table 6: Sample estimates against Horvitz–Thompson population estimates under the sampling design weights. Every dollar figure in this paper is a sample figure; the fleet is markedly more intermittent than the sample.

Quantity	Sample	Population
Mean cost per SKU (Chronos)	\$2,097	\$434
Demand-active SKU share	38.6 %	14.8 %
Holding-dominated SKU share	3.2 %	6.0 %
Cost share in holding-dominated	82.3 %	62.1 %
Mean within-SKU τ (demand-active)	+0.387	+0.436
$\rho(\text{MAE, cost})$, holding-dominated	+0.591	+0.991
$\rho(\text{MAE, cost})$, stockout-dominated	−0.345	−0.882
$\rho(\text{MAE, cost})$, pooled	+0.018	+0.109
Deployed gap	−50.7 %	−52.1 %

Every dollar figure in this paper is therefore a sample figure, and the fleet is far more intermittent than the sample: weighted, only 14.8 % of SKUs see demand in the test window. Chronos remains the cheapest model under weighting and the deployed gap is unchanged in direction and size.

The regime contrast sharpens rather than dissolving, which is convenient enough that it deserves scrutiny. A weighted correlation can be one heavy stratum wearing a large weight, so we decomposed it. The holding-dominated result is robust: it is positive within every major stratum (+0.845, +0.864, +0.809, +0.864 for the four largest by weight) and leave-one-stratum-out never takes it below +0.973. The stockout-dominated result is not. It falls from -0.882 to -0.182 when the Intermittent $\times$ C cell is removed, and the within-stratum values have no consistent sign, running from -0.709 to $+0.936$. That cell is 68 % of the fleet by weight, so -0.882 is a true statement about this population and a statement about one stratum at the same time.

So the contrast survives both checks and the magnitude of its negative half does not. Accuracy predicts cost where holding cost dominates, in the sample and in the population alike. Where stockouts dominate it does not, and how negative the relationship looks depends on the stratum. Kish’s effective sample size under these weights is 3,127 overall and 162 in the holding-dominated group, so the weighted intervals are wide by construction, and we report them alongside the unweighted ones rather than in place of them.

6 Discussion

The results cohere around one point: “does forecast accuracy predict inventory cost” is not a well-posed question until two things are fixed, the cost regime and the level of aggregation.

Fix the regime and the answer is clean. Where holding cost dominates, accuracy predicts cost strongly and the most accurate model is the cheapest one. Where stockout cost dominates, accuracy carries no fleet-level signal and can point the wrong way. That is Theodorou et al.’s finding, and it survives transfer to a different industry, a much more intermittent demand process, a different policy class and a deployed baseline. The transfer is the contribution; the sign reversal is the sharpening.

Fix the aggregation level and a second apparent contradiction dissolves. A positive within-SKU τ and a null fleet-level ρ on the same data are not in tension. They are different estimands, and the fleet-level one is closer to the decision a planner actually makes.

For practice this yields two cheap diagnostics, both computable before any model is fitted. The first is on the evaluation window: compute the share of SKUs whose test window has a zero median, since that share determines whether MAE can separate forecast quality from forecast level at all. At 77.3 % it cannot; at the 10.1 % we measure on M5 it largely can. The second is on the cost structure: compute $q^* = C_u / (C_u + C_o)$ per SKU. Where it sits near 0.5, MAE is aligned with the decision; where it does not, and here the median is 0.981, the metric targets the wrong fractile and its relationship with cost takes whatever sign the cost structure imposes. If a distributional forecast is available, score it at q^* under a policy that actually consumes the distribution, and not with a generic proper scoring rule, which we find performs worse than MAE. Both diagnostics point the same way as our third result: the safety-stock parameterisation separates the deployed policy from every reasonable model-driven alternative by far more than the forecasters separate from each other.

7 Limitations

1. **Four-quarter test horizon**, forced by the data vintage. Short for intermittent demand.
2. **One operator, one segment**. External validity is unestablished; this is a replication on one new domain, not a general law.

3. **The design weights are large and the effective sample is small.** Section 5.9 reports Horvitz–Thompson estimates alongside the unweighted ones, so fleet claims are no longer read off a raw sample mean. But w_h reaches 96, Kish’s effective n is 3,127, and in the holding-dominated group it is 162. Weighted intervals are correspondingly wide, and the weighted stockout-dominated correlation is carried by a single stratum.
4. **Cost concentration, and what it costs us in power.** 3.2 % of SKUs carry 82.3 % of simulated cost, so sample-mean cost is effectively a statement about a small high-value subset. Under a paired bootstrap only one of ten models is distinguishable from the cheapest, so the cost ranking should be read as descriptive. A design that trims or stratifies the tail would have more power, at the cost of no longer describing actual spend.
5. **Simulator not validated against realised stockouts**, because the data contain none. It under-stocks by roughly $2\times$ against the recorded snapshot.
6. **Lost-sales rather than backorder dynamics**, which suits consumable spares but not repairable or critical items where demand genuinely queues.
7. **Saturated start and a short window interact.** With on-hand initialised at S , 83.0 % of SKUs never reorder within the four test quarters, so the comparison largely measures holding cost on an initial position. A longer window, or a zero start, would exercise the replenishment logic more.
8. **Cost is 90.5 % holding**, so the cost ranking is close to a bias ranking and does not isolate the value of a forecast’s shape.
9. **Assumed cost parameters**, swept but not operator-supplied. The regime crossover at \$1,600 is a consequence of the assumed \$100 stockout penalty; a different penalty moves the boundary, though not the qualitative result.
10. **Model-agnostic σ_D** , which isolates point-forecast quality but does not test forecaster-specific uncertainty.
11. **Untuned baselines.** ZIP and HNB are intercept-only and share a point forecast; LightGBM is recursive multi-step; Chronos is zero-shot.
12. **The parameters are inherited, not fitted.** The information-set asymmetry that used to sit here is fixed in Section 5.9. What remains is that the smoothing constants for SBA, Croston, TSB, SES and the aggregate methods come from the operator’s deployed configuration rather than being optimised on the training window, so these methods are evaluated as deployed rather than at their best.
13. **Coarse granularity:** quarterly review, integer-quarter lead times, and a cycle-service z applied to criticality tiers specified as availability targets.
14. **The distributional comparison rests on four models.** Only ZIP, Hurdle-NB, LightGBM and Chronos emit quantiles, so the fleet-level version of Section 5.7’s metric comparison has four points and is not interpreted.
15. **The M5 comparison is a metric test, not a cost test.** We reuse published per-series accuracy and bias and recover forecast level from bias; no inventory simulation is run on M5, so Section 5.8 establishes the confound’s prevalence there but not its cost consequences.

8 Conclusion

On 15,348 SKUs of real maritime spare-parts data, the relationship between forecast accuracy and simulated inventory cost is conditional in exactly the way Theodorou, Spiliotis and Assimakopoulos (2025) predicted from M5 retail data, and more sharply. Where holding cost dominates, fleet-level rank correlation is $+0.591$; where stockout cost dominates it is -0.345 ; pooled across a population that is 96.8 % stockout-dominated it is $+0.018$. But the pattern is a confound rather than a signal. MAE is minimised at the median of realised demand, zero for 77.3 % of these SKUs, so MAE-rank is level-rank by identity, with per-SKU correlation median $+1.000$. The crossover follows analytically: the critical fractile $q^* = C_u/(C_u + C_o)$ equals the 0.5 fractile MAE targets at exactly \$1,600. Scoring at q^* instead does not help, because it inherits the confound with the sign reversed; only a policy that consumes the whole predictive distribution opens a usable channel, and then only when scored at the fractile the decision reads. On public M5 data the confound is mild (median $+0.194$) because the analytic condition holds for 10.1 % of series rather than 77.3 %, so its severity is a measurable property of the evaluation window rather than of intermittency. The answer also depends on aggregation: the same data give a moderate positive within-SKU τ of $+0.387$ and a null fleet-level correlation. A zero-shot foundation model is both the most accurate and the cheapest forecaster in the set, and ten of eleven model-driven policies beat the operator’s deployed parameterisation in direction. Censoring of the most recent quarters, though real and severe in this feed, does not reorder the forecasters. The regime contrast survives both design checks. Equalising the information sets across forecasters leaves it at $+0.591$, and carrying the sampling design weights raises it to $+0.991$, positive within every major stratum. Its negative half travels less well. That one collapses from -0.882 to -0.182 once the stratum holding 68 % of the fleet is removed, so what we claim is the contrast and not the magnitude on the stockout-dominated side.

Reproducibility and analysis note

The full pipeline (anonymisation, sampling, all eleven forecasters, the (s, S) simulator and every robustness check) is released as code, and the anonymised data are released with it. Results were produced under a claims-verification script that cross-checks every number in this paper against the analysis ledgers; it is part of the release.

This paper went through an adversarial review that changed its conclusions, and the corrections are worth recording because each one is a plausible failure mode for a cost-aware forecasting study.

1. **A predictive-quantile composition bug.** LightGBM’s quantiles were composed as the conditional-size quantile multiplied by an indicator that the occurrence probability exceeded 0.5. That collapsed the entire predictive distribution to a point mass at zero for two thirds of the sample and left the rest on an un-remapped conditional quantile. Corrected to the hurdle mixture, LightGBM’s Q99 coverage moved from 0.941 to 0.995 and its fill-matched cost outcome from $+204$ % to -64.4 %. An earlier draft reported the buggy figure as evidence that machine-learning models have tails too thin for inventory use.
2. **Monte Carlo sample size masquerading as miscalibration.** Chronos was sampled 30 times while ZIP and HNB were sampled 1,000 times, which biases the empirical 0.99 quantile downward. At 1,000 seeded draws its Q99 coverage moved from 0.939 to 0.961 and it reached target fill at $\kappa = 3.83$ rather than failing at the search bound.
3. **Non-deterministic training, in two layers.** LightGBM produced MAE of 8.526, 8.555 and

8.987 on three runs at the same seed. The first layer was the model: its stochastic parameters were unseeded and LightGBM is non-deterministic when multithreaded, so we pinned the seeds, `deterministic`, `force_row_wise` and the thread count. That was not sufficient. The second layer was the data: Polars `unique()` does not guarantee row order, and LightGBM’s bagging samples rows by position, so a reshuffled frame still produced a different model. Only after sorting the analysis frame to a canonical order did two separate processes agree bit-for-bit. Every number here is from the sorted pipeline; the regime correlation in particular moved from +0.673 ($p = 0.023$) to +0.591 ($p = 0.056$), which is why we do not lean on it.

4. **An unsupported headline.** An earlier draft claimed censoring flipped the cheapest forecaster to the most expensive. No artifact supported it, and the controlled experiment in Section 5.6 shows it is false.
5. **An over-strong dominance claim.** An earlier draft stated the deployed policy was beaten by every model-driven policy. LightGBM does not beat it.
6. **A vacuous robustness check.** An earlier draft offered MASE as corroboration of the MAE-based rank statistic. MASE is a positive per-SKU rescaling of MAE, so the statistic is identical by construction.
7. **A statistic with too little power.** The earliest version measured the accuracy-cost relationship as a Spearman correlation over eleven forecasters, whose bootstrap interval spanned most of $[-1, 1]$.
8. **An unfair comparison.** SBA, Croston and TSB were scored on fifteen training quarters against sixteen for the other eight models, because their state update lags the observation by one period. Section 5.9 equalises it. All three improve and nothing else moves.
9. **A bootstrap interval quoted more precisely than it was estimated.** At $B = 2,000$ the holding-dominated lower bound ranged from +0.409 to +0.545 across seeds, and an earlier draft quoted +0.527 from one of them. B is now 20,000, the interval is quoted to two decimals, and the stable statistic, the bootstrap share of replicates at or below zero (0.0014), is reported with it.
10. **Fleet claims read off an equal-allocation sample.** Design weights reach 96 : 1 here, and the unweighted mean cost is roughly six times the population estimate. Section 5.9 reports both.

One discrepancy remains unresolved: the Hurdle-NB fill-matched delta is +2.1 % in the main analysis script and -0.5 % in the robustness script, on the same window and nominal configuration. We report the main-script value and flag the inconsistency rather than choosing silently.

References

- Ansari, A. F., et al. (2024). *Chronos: Learning the Language of Time Series*. Transactions on Machine Learning Research. arXiv:2403.07815.
- Bandara, K., Bergmeir, C., & Smyl, S. (2020). Forecasting across time series databases using recurrent neural networks on groups of similar series: a clustering approach. *Expert Systems with Applications*, 140, 112896.
- Croston, J. D. (1972). Forecasting and stock control for intermittent demands. *Operational Research Quarterly*, 23(3), 289–303.

- Gneiting, T., Balabdaoui, F., & Raftery, A. E. (2007). Probabilistic forecasts, calibration and sharpness. *JRSS B*, 69(2), 243–268.
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688.
- Ke, G., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS* 30.
- Kolassa, S. (2016). Evaluating predictive count data distributions in retail sales forecasting. *International Journal of Forecasting*, 32(3), 788–803.
- Kourentzes, N. (2014). On intermittent demand model optimisation and selection. *International Journal of Production Economics*, 156, 180–190.
- Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). The M5 accuracy competition: results, findings, and conclusions. *International Journal of Forecasting*, 38(4), 1346–1364.
- Nikolopoulos, K., Syntetos, A. A., Boylan, J. E., Petropoulos, F., & Assimakopoulos, V. (2011). An aggregate–disaggregate intermittent demand approach (ADIDA) to forecasting: an empirical proposition and analysis. *JORS*, 62(3), 544–554.
- Petropoulos, F., Wang, X., & Disney, S. M. (2019). The inventory performance of forecasting methods: evidence from the M3 competition data. *International Journal of Forecasting*, 35(1), 251–265.
- Prak, D., & Teunter, R. (2019). A general method for addressing forecasting uncertainty in inventory models. *International Journal of Forecasting*, 35(1), 224–238.
- Salinas, D., Flunkert, V., Gasthaus, J., & Januschowski, T. (2020). DeepAR: probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3), 1181–1191.
- Silver, E. A., Pyke, D. F., & Thomas, D. J. (2017). *Inventory and Production Management in Supply Chains* (4th ed.). CRC Press.
- Snyder, L. V., & Shen, Z.-J. M. (2019). *Fundamentals of Supply Chain Theory* (2nd ed.). Wiley.
- Syntetos, A. A., & Boylan, J. E. (2005). The accuracy of intermittent demand estimates. *International Journal of Forecasting*, 21(2), 303–314.
- Syntetos, A. A., & Boylan, J. E. (2006). On the stock control performance of intermittent demand estimators. *International Journal of Production Economics*, 103(1), 36–47.
- Teunter, R. H., Babai, M. Z., & Syntetos, A. A. (2010). ABC classification: service levels and inventory costs. *Production and Operations Management*, 19(3), 343–352.
- Teunter, R. H., Syntetos, A. A., & Babai, M. Z. (2011). Intermittent demand: linking forecasting to inventory obsolescence. *European Journal of Operational Research*, 214(3), 606–615.
- Theodorou, E., Spiliotis, E., & Assimakopoulos, V. (2025). Forecast accuracy and inventory performance: insights on their relationship from the M5 competition data. *European Journal of Operational Research*, 322(2), 414–426. doi:10.1016/j.ejor.2024.12.033.
- Vandeput, N. (2020). *Inventory Optimization: Models and Simulations*. De Gruyter.